

GROUP ASSIGNMENT

DATA PREPROCESSING

@ AIMS Senegal, 2025

OVERVIEW

- This group assignment **invites you to collaborate with your colleagues** to explore data and extract insights.
- Each group will be given a dataset to conduct an exploratory analysis and report on their findings.



GROUP I: CUSTOMER ANALYTICS FOR NIGERIABANK

NigeriaBank is one of Nigeria's leading commercial banks with over 500 branches across the country, serving millions of customers across the country. The bank has been experiencing challenges in understanding customer behavior, optimizing branch performance, and managing loan portfolios effectively.

The executive management team has identified several critical business questions. Your team has been hired to work on analyzing the bank's customer database to uncover actionable insights that can enhance customer relationship management, improve branch operations, and optimize lending decisions.

You have been provided access to the bank's customer database in a CSV file named `nigerian_banking_data.csv` containing comprehensive customer information.



Project 2: Agricultural Intelligence for Kenya Agricultural Research Institute (KARI)

GROUP 1: CUSTOMER ANALYTICS FOR NIGERIABANK

Kenya Agricultural Research Institute (KARI) is the leading agricultural research organization in Kenya, responsible for developing and promoting agricultural technologies to enhance food security and farmer livelihoods across the nation. Agriculture is the backbone of Kenya's economy, employing over 40% of the population and contributing significantly to GDP.

KARI has been collecting comprehensive data on crop production, weather patterns, and market prices across different counties in Kenya over several years. However, the institute faces critical challenges and the Ministry of Agriculture and farmer cooperatives are requesting data-driven expertise to leverage their datasets so that the insights extracted from it can lead to recommendations to improve agricultural productivity and farmer incomes.

Your team has been hired as **Agricultural Data Analyst** at KARI, you have been tasked with analyzing time series data to uncover insights that can improve agricultural planning, enhance food security, and support farmer decision-making.



Project 03: Mining Operations Intelligence for Minerals Council South Africa

OVERVIEW

- **Minerals Council South Africa** is the leading organization representing the country's mining industry, which is a cornerstone of South Africa's economy. The mining sector contributes significantly to GDP, exports, and employment, with hundreds of thousands of workers across the nation.
 - The executive committee needs data-driven insights to develop industry policies, improve safety standards, and enhance operational efficiency. **Your team has been hired as. Mining Data Analyst** at the Minerals Council, you have been tasked with analyzing operational data to uncover insights that can improve mining efficiency, enhance safety protocols, and support policy development.
 - The client is particularly interested in understanding production patterns, safety performance, regional variations, and economic indicators, as this information is crucial for developing evidence-based mining policies and operational guidelines.
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Project 04: West African Agricultural Market Intelligence



ANALYSIS AND RESEARCH REQUIREMENTS

West African Agricultural Markets Consortium (WAAMC) is a regional organization that monitors and supports agricultural trade across West Africa. The region's agricultural markets are crucial for food security, economic stability, and the livelihoods of millions of farmers and traders across countries including Nigeria, Ghana, Senegal, Burkina Faso, and other.

The consortium has been collecting comprehensive data on crop prices, trading volumes, and market conditions across major agricultural markets in West Africa. Regional governments and farmer cooperatives are requesting evidence-based insights to improve market efficiency and farmer incomes.

As a **Market Intelligence Analyst** Team at WAAMC, you have been tasked with analyzing agricultural market data to uncover insights that can improve market transparency, support farmer decision-making, and enhance food security across the region.

Project 05: Mobile Money Transaction Intelligence for East African Mobile Payments Consortium



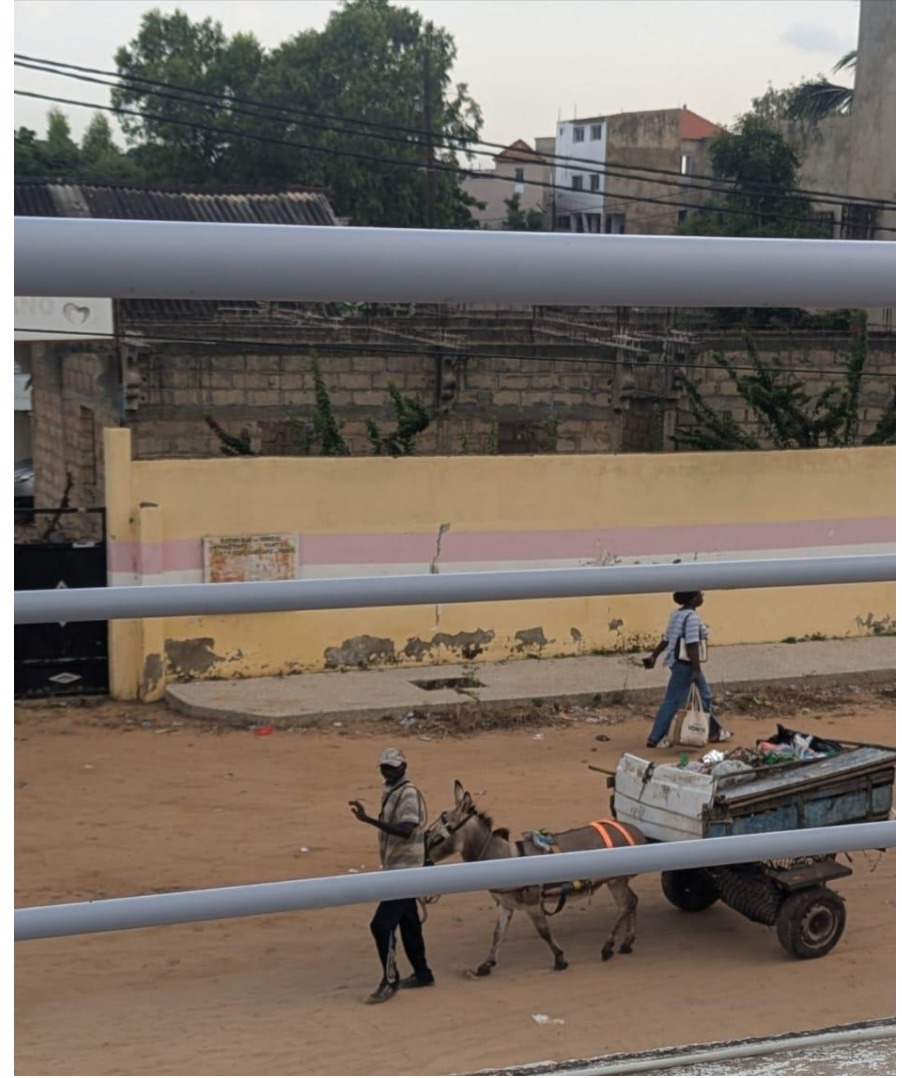
ANALYSIS AND RESEARCH REQUIREMENTS

East African Mobile Payments Consortium (EAMPC) is a regional organization that coordinates and analyzes mobile money services across East Africa. Mobile money has revolutionized financial services in Africa, with Kenya's M-Pesa leading the way, followed by similar services in Tanzania and Uganda.

The consortium has been collecting comprehensive transaction data across Kenya, Tanzania, and Uganda. Regulators, service providers, and development organizations are requesting insights to improve service delivery and financial inclusion.

Your team is operating as **Financial Data Analysts** at EAMPC and you have been tasked with analyzing mobile money transaction patterns to uncover insights that can optimize system capacity, improve service delivery, and support financial inclusion initiatives.

Project 06: Inventory Intelligence for SenLogistics



ANALYSIS AND RESEARCH REQUIREMENTS

SenLogistics is a prominent logistics and distribution company operating multiple warehouses across various regions in Senegal, including Dakar, Kaolack, Ziguinchor, and Tambacounda.

The company serves as a critical link in the supply chain, managing inventory for electronics, furniture, toys, and other consumer goods across the country. You have been provided access to the company's warehouse database in a CSV file named `SenLogistics_warehouses.csv` containing comprehensive inventory information across all warehouse locations.

As a newly hired **Supply Chain Data Analysts** at SenLogistics, you have been tasked with analyzing the warehouse data to uncover insights that can enhance the overall inventory and supplier management process.



Project 07: IT Infrastructure Intelligence for AIMS Senegal

ANALYSIS AND RESEARCH REQUIREMENTS

African Institute for Mathematical Sciences (AIMS) Senegal is a pan-African center of excellence in mathematical sciences, hosting students, researchers, and faculty from across Africa. The campus operates 24/7 with intensive computational needs, research activities, and administrative operations.

The campus infrastructure includes 78 devices across 6 departments, generating hourly operational logs. The administration needs comprehensive insights to

- Reduce operational costs through better resource management
- Improve service reliability for students and researchers
- Plan equipment procurement and replacement strategically
- Optimize energy consumption
- Enhance security and monitoring

As **Data Analysts** within the IT Department at AIMS Senegal, your team has been tasked with performing comprehensive log analysis across all equipment and departments on campus to support data-driven decision-making.

ANALYSIS AND RESEARCH REQUIREMENTS

I. Research Local Context

- Briefly research how the selected data variables reflect the **local realities** of the two cities.
- Use online sources, articles, or personal knowledge to describe how these variables influence or relate to life in the chosen cities.

2. Exploratory Data Analysis (EDA)

- Perform **data cleaning** where necessary to ensure data quality. Report the different cleaning steps
 - Create **visuals** (e.g., charts, plots) to uncover trends, relationships, or patterns in the data.
 - Conduct a **correlation analysis** (if applicable) to explore relationships between the variables.
 - You could go beyond the techniques we covered in class—feel free to use other EDA techniques to enhance your analysis.
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SUBMISSION PACKAGE

1. **Jupyter Notebook File:** Compile your work in a Jupyter Notebook so that we can make sense of your workflow. The notebook should include:

- A brief introduction to your cities and dataset
- Data cleaning steps
- Visualizations with insights
- Summary of key findings

4. **PowerPoint Presentation**

- Create a power presentation/beamer that highlighting your findings.
 - Include key visuals, insights from your research, and a summary of your analysis.
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GUIDELINES

I. Submission Guidelines

- Jupyter Notebook: Submit the notebook
- Powerpoint Presentation (PDF): Be ready to present your findings in **12 minutes** on the final day.

2. Assessment Criteria

- **Collaboration and teamwork:** How effectively the group worked together. EVERYONE MUST PRESENT
 - **Data analysis and research quality:** The depth of your research, EDA, and insights.
 - **Visualizations:** The clarity and effectiveness of your visuals.
 - **Presentation:** Your ability to clearly communicate your findings.
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GROUP FORMATION : 6 GROUPS 3 TOPICS

PROJECT 1

- Okere Rafiatou
- Ndeye Binta Ndiaye
- MUNGUAGANZE BWIRACHIZA
CHRISTIAN
- OGOUNCHI Géraud Badélé

PROJECT 2

- Gaolatlhe Angelah Kgato
- LIEUMO NJOPNU MARCELLE ALIDA
- Harilova Julianna Ramina

PROJECT 3

- LEKOUNDA NGOLO Mardochet Gédéon
- Charlotte Chepngeno Sigei
- Adama Telly Ba

PROJECT 4

- Ahmed Souleymane Sow
- Sangura JOEL Nyongesa
- Marie Paul Basse

PROJECT 5

- Ndeye Khady Wade
- Soukeye Toure
- Maimouna Ndoeye

PROJECT 7

- Ansoumane Conde
- TSEMO DANIELLE
- SANE Fatoumata
- Moute Jean Baptiste

PROJECT 6

- Akakpo Afandoda Adjo Sara
 - Wandiya James
 - MAIRAME SAMBA NIANG
 - Daouda Sene
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THANK YOU !!!

